

Verify each identity.

$$13) \frac{\sin 2x}{\cos x} = 2\sin x$$

$$14) \frac{2\sin x \cos x}{\cos 2x} = \tan 2x$$

$$15) \frac{2\sin x \cos x}{\csc^2 x - 1} = \tan^2 x \sin 2x$$

$$16) \frac{\sin 2x}{\cot^2 x (1 + \cos 2x)} = \tan^3 x$$